Route G-R11: One Preparation and One Record

Energy-compatible relative readout; explicit apparatus boundary

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Abstract

A relative-phase record can be written without changing the full system energy distribution. We construct an explicit 72-dimensional unitary, retain failed preparations, and calculate the measurement disturbance. This adds a collective apparatus coupling to the unchanged G-R10 system; it does not derive that coupling from the vertex contact. The same gate erases its record if left active. Energy compatibility therefore closes one finite instrument-existence question, not autonomous time generation.

1 Unchanged system; only the relative reading is recorded

In dimensionless units, keep the two-qutrit Hamiltonian at $\theta = \pi/2$:

$$L = (2-\sqrt{3})I + \sqrt{3}N, \qquad N = \mathrm{diag}(0,1,2), \tag{1}$$

$$H = L \otimes I + I \otimes L + \gamma P_v \otimes P_v, \quad v = (1,1,1)^T/\sqrt{3}, \quad P_v = vv^\dagger. \tag{2}$$

The inherited fixtures are $\gamma = 0.09, 0.03, 0.01$, not physical fits. Put $D = (|D_+\rangle, |D_-\rangle)$, $P = DD^\dagger$, $Q = I - P$, where

$$|D_+\rangle = \frac{|02\rangle - 2|11\rangle + |20\rangle}{\sqrt{6}}, \qquad |D_-\rangle = \frac{|02\rangle - |20\rangle}{\sqrt{2}}, \qquad HP = 4P. \tag{3}$$

These states are annihilated by the contact. For G-R10's fixed effects $E_r = w_r w_r^\dagger/7$, $w_r = (1, e^{\mathrm{i}\phi_r}, e^{2\mathrm{i}\phi_r})^T$, $\phi_r = 2\pi r/7$, define $\bar{F}_{rs} = \sum_E \Pi_E (E_r \otimes E_s)\Pi_E$. This energy averaging is an operator definition, not a supplied clock. For $d = (s-r) \bmod 7$, $\delta_d = 2\pi d/7$, direct projection gives

$$D^\dagger(w_r \otimes w_s) = e^{\mathrm{i}(\phi_r+\phi_s)} g_d, \qquad g_d = \begin{pmatrix} \sqrt{2/3}(\cos\delta_d - 1) \\ \mathrm{i}\sqrt{2}\sin\delta_d \end{pmatrix}, \tag{4}$$

$$C_d = D\frac{g_d g_d^\dagger}{7}D^\dagger, \qquad P\bar{F}_{r,r+d}P = \frac{C_d}{7}, \quad \sum_{d=0}^{6} C_d = P, \quad C_0 = 0. \tag{5}$$

Thus r is uniformly redundant for *every* state in P. The reduced joint Lüders instrument is also preserved by discarding r, since $\sqrt{C_d/7} = \sqrt{C_d}/\sqrt{7}$. This is not the instrument of separate local A/B phase measurements, nor a pair of absolute timestamps.

2 A complete instrument and an explicit conserving unitary

Include failure rather than presuming a free successful preparation:

$$K_\perp = Q, \qquad K_d = \sqrt{C_d} \quad (d = 1, \ldots, 6), \qquad \sum_\ell K_\ell^\dagger K_\ell = I, \qquad [H, K_\ell] = 0. \tag{6}$$

For $d \neq 0$ the rank-one square root is elementary:

$$K_d = D \frac{g_d g_d^\dagger / 7}{\sqrt{g_d^\dagger g_d / 7}} D^\dagger. \tag{7}$$

The dark projection and relative readout can therefore be one operation. It prepares a dark state on success, but not an undisturbed D_+.

Use one memory with eight orthogonal, energetically degenerate labels: $|b\rangle, |\perp\rangle, |1\rangle, \ldots, |6\rangle$ and $H_M = 0$. The initially pure blank b is distinct from failure; no hidden epoch bit supplies that distinction. Eight is a minimum only for these seven distinguishable outputs plus a separate blank in this register scheme. Define the 72×9 embeddings

$$J = I \otimes |b\rangle, \qquad W = Q \otimes |\perp\rangle + \sum_{d=1}^{6} K_d \otimes |d\rangle. \tag{8}$$

Equation (6) gives $J^\dagger J = W^\dagger W = I$ and $J^\dagger W = 0$. With $P_J = JJ^\dagger$, $P_W = WW^\dagger$, construct

$$A = WJ^\dagger - JW^\dagger, \quad G = \mathrm{i}A = G^\dagger, \quad A^2 = -(P_J + P_W), \tag{9}$$

$$U = I - P_J - P_W + A = \exp[-\mathrm{i}(\pi/2)G]. \tag{10}$$

Indeed, $UJ = W$, $UW = -J$, and U is the identity on the orthogonal complement. These relations prove unitarity without an unspecified extension. Since $H_{SM}J = JH$ and $H_{SM}W = WH$, where $H_{SM} = H \otimes I_M$, they also prove

$$[G, H_{SM}] = [U, H_{SM}] = 0. \tag{11}$$

Thus the instrument preserves the entire distribution of H, not just its mean. Measuring a memory projector produces the corresponding Kraus branch. Before that measurement the joint state need not be classical.

What has been added. G is a new *collective* system–memory interaction, not a consequence of $\gamma P_v \otimes P_v$ or a demonstrated local hardware coupling. For a controlled Hamiltonian $H_{SM} + \kappa(\tau)G$ and pulse area $a = \int \kappa(\tau)d\tau$, the interaction-picture gate is $e^{-\mathrm{i}aG}$. The physical propagator also contains $e^{-\mathrm{i}H_{SM}\tau}$. This factor leaves all record probabilities and energy statistics below unchanged, but must be retained for subsequent coherent experiments. Here τ and pulse control are assumed, not a newly derived time.

3 All outcomes, their energies, and the measurement disturbance

For the declared input $|11\rangle$,

$$P|11\rangle = -\sqrt{2/3}|D_+\rangle, \quad Q|11\rangle = |B\rangle/\sqrt{3}, \quad |B\rangle = (|02\rangle + |11\rangle + |20\rangle)/\sqrt{3}. \tag{12}$$

The failure probability is 1/3; success totals 2/3. Conditional on success the relative record has the normalized distribution

$$q_d = \langle D_+|C_d|D_+\rangle = \frac{2}{21}(1-\cos\delta_d)^2, \qquad p_\perp = \frac{1}{3}, \quad p_d = \frac{2}{3}q_d. \tag{13}$$

The finite Fourier sums give $\sum_d \cos\delta_d = 0$ and $\sum_d \cos^2\delta_d = 7/2$, proving normalization. Numerically:

Record	failure	$d = 1, 6$ (each)	$d = 2, 5$ / $d = 3, 4$ (each)
Probability	0.333333333	0.009000630	0.094892536 / 0.229440167

The complete energy ledger, including rejected preparations, is

$$E_{\text{in}} = 4 + \frac{\gamma}{9}, \quad E_S = 4, \quad E_\perp = 4 + \frac{\gamma}{3}, \quad \frac{2}{3}E_S + \frac{1}{3}E_\perp = E_{\text{in}}. \tag{14}$$

At $\gamma = 0.09$ these are 4.01, 4, and 4.03. B is the failure state for U in the interaction picture, not generally an eigenstate of H; its free evolution changes neither the failure probability nor its energy distribution. Success lies in the stationary eigenspace P.

Non-negligible backreaction. Each successful record yields a pure dark state proportional to Dg_d. If d is unread after success, the system becomes exactly

$$\rho'_S = \sum_{d=1}^{6} K_d|D_+\rangle\langle D_+|K_d = \frac{57}{82}|D_+\rangle\langle D_+| + \frac{25}{82}|D_-\rangle\langle D_-|. \tag{15}$$

This is an analytic ratio, not a decimal fit. To verify it, put $c_d = \cos(2\pi d/7)$. Pairing d with $7-d$ cancels the off-diagonal term. The D_+ weight is

$$F = \frac{1}{21}\sum_{d=0}^{6}\frac{(1-c_d)^3}{2+c_d} = -\frac{9}{2} + \frac{9}{7}\sum_{d=0}^{6}\frac{1}{2+c_d}, \tag{16}$$

$$\sum_d \frac{1}{2+c_d} = -\frac{T_7'(-2)}{T_7(-2)-1} = \frac{497}{123}, \qquad F = \frac{57}{82}. \tag{17}$$

Here $T_7(z) - 1$ has roots c_d counted with multiplicity; $T_7(-2) = -5042$, $T_7'(-2) = 20377$. Every successful branch and their mixture retain G-R10's conditional orbit $\rho_{B|\phi} = e^{\mathrm{i}N_B\phi}\rho_{B|0}e^{-\mathrm{i}N_B\phi}$, but with a changed seed state. This does not mean D_+ survives unchanged.

4 Which phase reference is avoided, and why a record can disappear

An elementary covariance argument makes the resource boundary precise. Let an apparatus have Hamiltonian H_R, state σ_R, pointer Z_ℓ, and joint unitary V. Suppose

$$[\sigma_R, H_R] = [Z_\ell, H_R] = 0, \qquad [V, H \otimes I + I \otimes H_R] = 0. \tag{18}$$

Its outcome probability and induced effect are

$$p_\ell(\rho) = \mathrm{tr}[V(\rho \otimes \sigma_R)V^\dagger(I \otimes Z_\ell)] = \mathrm{tr}(\rho F_\ell). \tag{19}$$

Write $T_t = e^{-\mathrm{i}Ht}$ and $R_t = e^{-\mathrm{i}H_R t}$. Invariance of σ_R gives $(T_t \rho T_t^\dagger) \otimes \sigma_R = (T_t \otimes R_t)(\rho \otimes \sigma_R)(T_t \otimes R_t)^\dagger$. Commuting this joint translation through V and the pointer, then using trace cyclicity, proves

$$p_\ell(T_t \rho T_t^\dagger) = p_\ell(\rho) \quad \text{for every } \rho, \qquad [F_\ell, H] = 0. \tag{20}$$

The relative effects C_d meet this condition, while the bare local phase effects $E_r \otimes I$ generally do not. For example, the inherited fixtures all give $\|[H, E_0 \otimes I]\|_{\mathrm{F}} \simeq 1.484615$. Thus our explicit U needs no external *energy-time phase asymmetry* resource, but does not implement the separate local phase POVM using such invariant resources. Internal relative calibration, pure blank memory, and collective control are still inputs. This is a direct finite covariance proof, consistent with the broader WAY literature [1]; it is not a new general measurement theorem.

Record versus autonomous event. Using $AJ = W$, $AW = -J$, the continuously active gate obeys

$$e^{-\mathrm{i}aG} J = \cos a\, J + \sin a\, W, \qquad P_{\mathrm{blank}}(a) = \cos^2 a. \tag{21}$$

At $a = \pi/2$ a record exists. At $a = \pi$ the entire input is blank again, up to sign: $U^2 J = -J$. Including free system evolution changes that system vector but not its blank-memory status. Therefore a static energy-conserving gate is not, by itself, a durable autonomous recorder.

If the coupling is explicitly turned off after writing, subsequent $e^{-\mathrm{i}Ht} \otimes I_M$ preserves every memory label. Its value can be reread without rerunning the system POVM; the latter is nonprojective and need not yield the same d again. Actual isolation, noise and lifetime are not modeled. The premeasurement may also retain coherence between labels, so neither irreversible classicalization nor a universal arrow follows from the unitary. The missing physical choice is a trigger and decoupling mechanism, not a smaller numerical error.

5 Information ledger, verification and bounded conclusion

For a *classicalized* record the Shannon entropy, in bits, is

$$H_{\mathrm{rec}} = h_2(2/3) + \frac{2}{3}H(q) = 2.27003421247, \qquad H(q) = 2.02760756762, \tag{22}$$

$$H_{\mathrm{redundant}} = \frac{2}{3}\log_2 7 = 1.87156994804. \tag{23}$$

The second number is the per-attempt information cost of also saving the uniformly redundant common-origin label r. Removing it loses no dark-sector relative instrument, by Eq. (5); it does not remove failure or selectively discard unfavorable records.

For ideal thermal resetting of a degenerate classical register at temperature T, no accessible side information, and cyclic controls, the free-energy bound is [2]

$$W_{\mathrm{reset}} \geq k_B T \ln 2\, H_{\mathrm{rec}}. \tag{24}$$

This is a conditional reset bound, not a universal writing cost or an actual heat prediction. A coherent register, accessible system–memory correlations, a finite bath, or a noncyclic controller require their own ledger. No kelvin/joule calibration is supplied. $H_M = 0$ and conservation of bare energy do not make pure blank preparation, control, fabrication or resetting free. The apparatus pulse has not been embedded into an autonomous energy source.

Reproducibility and scope. The fixed card, executable verifier and machine-readable results are, respectively, under the repository's Route-G data, code and output paths:

```
data/PREPARATION_RECORD_CARD.json
code/verify_gr11_preparation_record.py
output/gr11_preparation_record.json
```

There are 120 passing stage checks: all 49 invariant effects compress exactly, the 72-dimensional unitary and pulse exponent agree, every energy spectral weight is conserved, failure and backreaction are retained, and continued coupling returns the memory to blank. G-R10's code, card and result hashes are pinned and unchanged. Numerical checks corroborate the identities; they do not substitute for the derivations or test real apparatus data. The independent G-R9b paired-channel side branch is retained, not imported into this interaction.

Decision. The bounded energy-compatible instrument-existence task closes here. One may keep this as an effective relational-clock measurement with explicit control assumptions. If the aim remains autonomous event/time generation, next choose a concrete physical trigger and self-decoupling mechanism and account for its energy and records. Do not add another abstract measurement prerequisite, bath, fitted time law or parameter scan as a substitute for that choice. This construction proves neither the source paper's E_d law nor emergent seconds, spacetime, particle masses, or any Route-F promotion.

References

[1] I. Marvian and R. W. Spekkens, *An information-theoretic account of the Wigner–Araki–Yanase theorem*, arXiv:1212.3378 (2012).

[2] D. Reeb and M. M. Wolf, *An improved Landauer Principle with finite-size corrections*, New J. Phys. **16**, 103011 (2014), arXiv:1306.4352.